

Abstract No.

Neurosurgical skull bone dust for refined cranioencephalic immune compartment access in glioblastoma

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Although immunotherapies have achieved notable clinical success in many oncological diseases, glioblastoma (GB), the most prevalent malignant brain tumor in adults, remains incurable and prone to relapse. Augmenting anti-tumor immunity is particularly challenging because effector immune cells are underrepresented within the tumor microenvironment. Our group recently identified tumor-reactive CD8+ effector/memory T cells in the adjacent skull bone marrow (Dobersalske et al., Nat Med, 2024), indicating a potential endogenous reservoir to enhance a potent anti-tumor immune response. Here, we evaluate methods for advanced T cell isolation from GB patients' skull bone, presenting a whole new protocol enabling direct extraction from bone dust obtained during neurosurgical craniotomy. Our aim is to recycle an organic remnant occurring during brain tumor-surgery to facilitate specimen collection as well as enable punctual guided and targeted approaches to the cranial anti-tumor T cell compartment in the skull of GB patients. In addition to the established protocol that enables the use of cranial bone dust as sample material for successful T cell isolation from the skull bone marrow, comparison of cell count, viability, expansion curves and T cell subpopulation compositions shows a direct correlation between surgical burr hole drilling instruments and sample quality in terms of T cell fractions that provides essential information of how to effectively access the cranial skull bone marrow. Hence, our protocol using bone dust for simple access to the immune compartment in the skull bone offers an opportunity to decipher immune cell patterns in the skull, to unlock novel routes for precise and optimized immune cell collection, to understand anti-tumoral functionality and thus to identify therapeutically relevant sites of the cranial immune cell reservoir against glioblastoma.



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